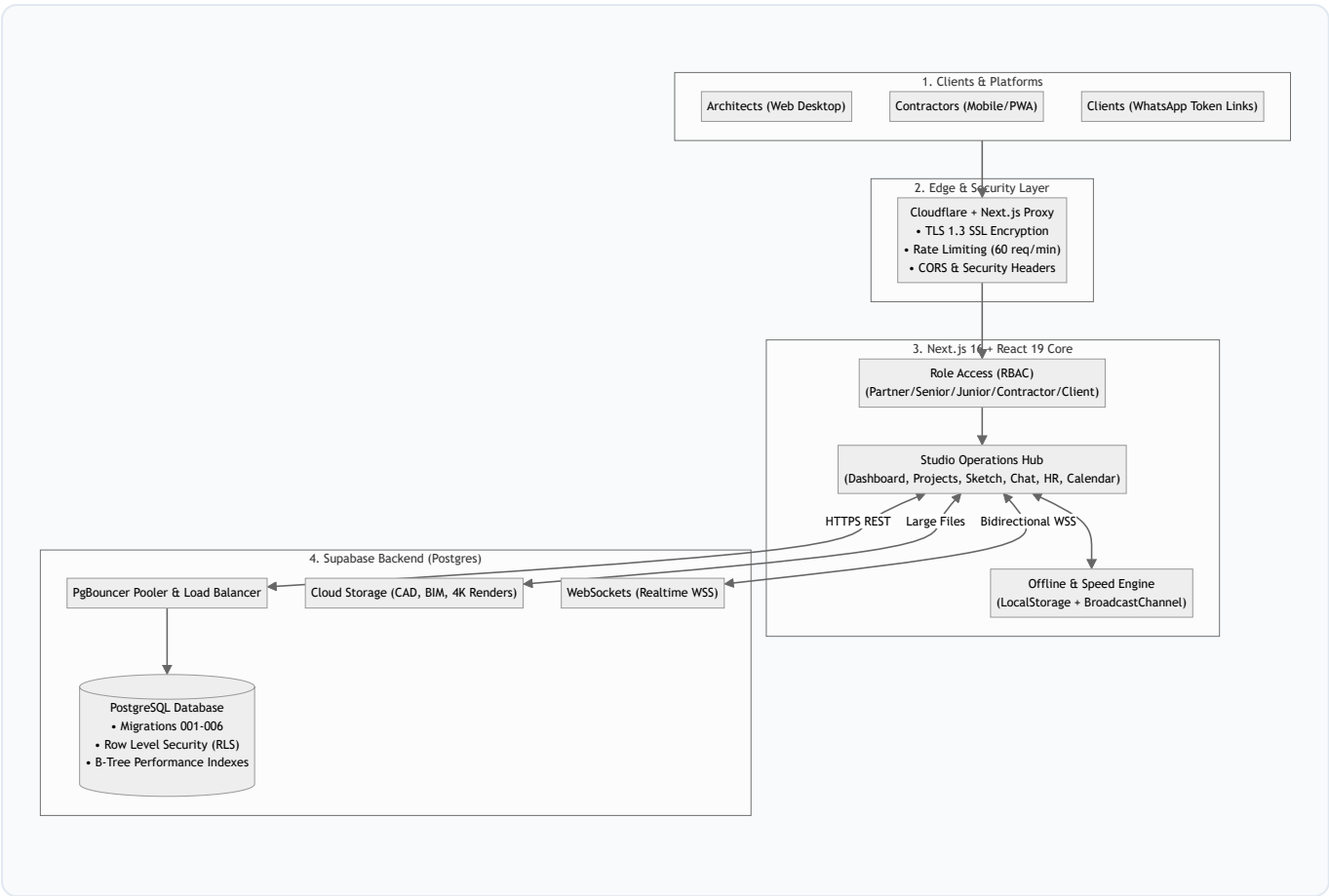


1. EXECUTIVE SUMMARY & VISION

Estudio Arkipelago is a unified, high-performance digital workspace and operations platform engineered specifically for architectural practices, structural engineers, construction contractors, and external clients. The platform eliminates fragmented communication channels by combining **interactive drawing annotation**, **construction administration (RFIs, Submittals, BP 344 compliance)**, **time tracking & HR ledgers**, and **tokenized client review portals** into a single reactive system.

2. END-TO-END SYSTEM ARCHITECTURE FLOWCHART



TECHNICAL SPECIFICATIONS

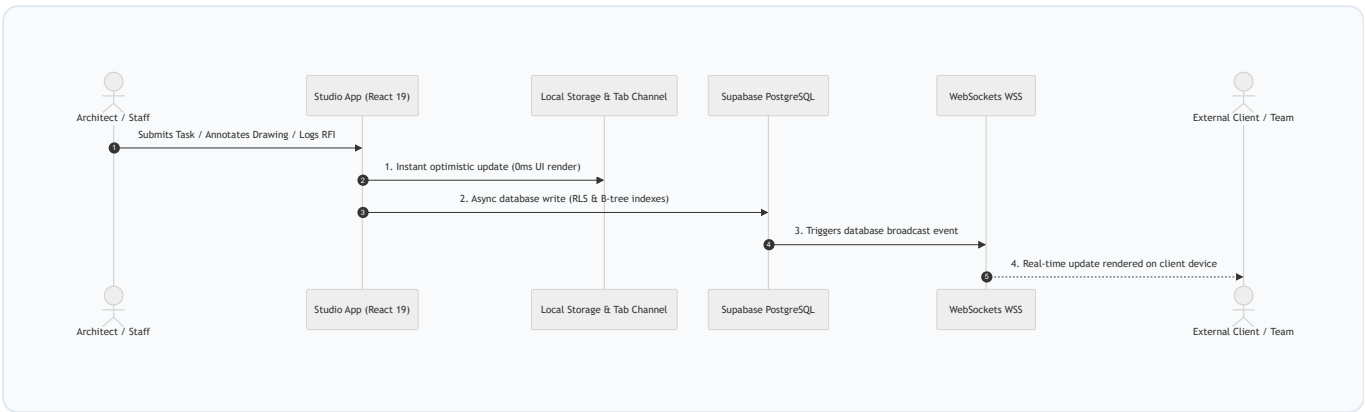
INFRASTRUCTURE, SECURITY, DATABASE & STATE
MANAGEMENT

Estudio Arkipelago
Architecture
Section: Technical Inventory

3. TECHNICAL INFRASTRUCTURE MATRIX

LAYER	TECHNOLOGIES USED	SYSTEM RESPONSIBILITIES & GUARANTEES	STATUS
Edge & Security	Cloudflare, Next.js Middleware, HSTS	TLS 1.3 HTTPS encryption, 60 req/min per IP rate limiting on <code>/api/*</code> , CORS protection, X-Frame-Options SAMEORIGIN.	ACTIVE
Application Core	Next.js 16.3 (Turbopack), React 19	17 active routes, responsive studio shell, interactive drawing canvas (<code>/sketch</code>), Construction Admin & BP 344 calculators.	ACTIVE
Offline & Caching	LocalStorage, BroadcastChannel	Oms instant UI first-paint, multi-tab state sync without network roundtrips, offline job site drafting resilience.	ACTIVE
Database & Storage	Supabase PostgreSQL, PgBouncer, Storage	6 version-controlled migrations (001-006), Row Level Security (RLS) data isolation, composite B-tree performance indexes.	ACTIVE
Real-Time Layer	WebSockets (WSS), Supabase Realtime	Sub-50ms live task board updates, typing indicators, studio feed reactions, cross-tab clock-in sync.	ACTIVE
CI/CD & DevOps	GitHub Actions, Playwright, Vercel	Automated 4-stage quality gate (TypeScript check → ESLint → Production Build → Playwright E2E browser tests).	ACTIVE
Client Review Portal	Tokenized Guest Authentication	Passwordless client approval links with drawing annotation pins and digital milestone sign-offs.	PHASE 1 READY

4. REAL-TIME DATA FLOW SEQUENCE

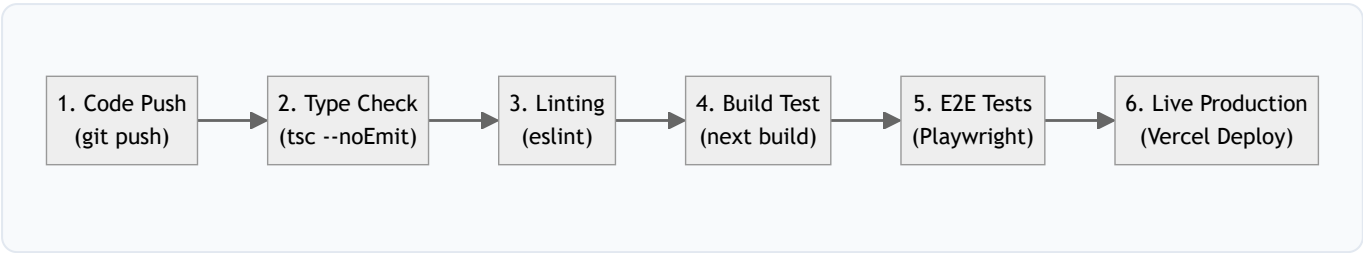


QUALITY ASSURANCE & DELIVERY ROADMAP

CI/CD PIPELINES, ROLE-BASED ACCESS & PHASED EXPANSION

Estudio Arkipelago
Architecture
Section: QA & Roadmap

5. AUTOMATED CI/CD & TESTING PIPELINE



6. ROLE-BASED ACCESS CONTROL (RBAC) HIERARCHY

ROLE	ACCESS PERMISSIONS & CAPABILITIES	RESTRICTED BOUNDARIES
Partner / Principal	Full studio oversight, fee proposals, contract billing in PHP (₱), HR approvals, global financial analytics.	None (Super-Admin).
Senior Architect	Project management, RFI reviews, submittal approvals, blueprint version control, task assignment.	Confidential studio HR complaint logs & master financials.
Junior Architect	Sketch drawing canvas, task deliverables, time tracking logbook, studio wall communication.	Contract pricing, HR admin, and project deletion.
Contractor / Site Engr.	Submission of RFIs, material submittal uploads, site progress photos for assigned project code.	Isolated strictly to assigned project; zero access to studio directory, HR, or other projects.
External Client Guest	Viewing project drawings, 3D renders, pin comment feedback, 1-click milestone approval.	Scoped strictly to tokenized review URL. Zero access to internal studio operations.

7. STRATEGIC DELIVERY ROADMAP

Phase 1 (Immediate Next Step): Client Review & Approval Portal (`/review/[token]`) allowing clients to approve drawing revisions via WhatsApp/Email links with a legal audit trail.

Phase 2: Multi-user collaborative drawing cursors inside `/sketch` using Supabase Realtime Broadcast.

Phase 3: AI-powered building code compliance scanner (BP 344 / Zoning Setbacks) and automated meeting minutes task extraction.